NAHID KARIM ANKUR

Software Engineer | Backend & Full-Stack Systems

+880 1962-856913 | nahidankur@gmail.com | Dhaka, Bangladesh

linkedin.com/in/nahidankur | github.com/nahidankur | nahidankur.github.io

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience developing, optimizing, and maintaining enterprise-grade backend services and full-stack applications. Proficient in Node.js, NestJS, Hapi.js, Express.js, Angular, PostgreSQL, Oracle, and MongoDB, with expertise in RESTful API development, database transaction management, query optimization, and production system reliability. Experienced in building scalable regulatory and financial systems, resolving complex production issues, implementing secure workflows, and establishing application monitoring solutions.

TECHNICAL SKILLS

Programming Languages:	JavaScript, TypeScript, C/C++, Python
Backend Development:	Node.js, NestJS, Hapi.js, Express.js, RESTful API Development, Backend Architecture
Frontend Development:	Angular, React, Next.js, NG-ZORRO, Tailwind CSS, Material UI, Bootstrap, HTML5, CSS3
Databases & Data Management:	PostgreSQL, Oracle, MongoDB, SQL, Relational Database Modeling, Transaction Management, Query Optimization
Monitoring & DevOps:	Grafana, Prometheus, Loki, Docker, CI/CD Pipelines, Linux, Git, Environment Deployment
Security & System Reliability:	Role-Based Access Control (RBAC), CAPTCHA Integration, Data Integrity, Production Troubleshooting, Application Monitoring
AI & Agentic Development:	Claude Code, GitHub Copilot, Antigravity, Agentic Workflows, AI-Assisted Coding
Tools & Reporting:	Postman, JasperReports

PROFESSIONAL EXPERIENCE

Celloscope Ltd. | ***Software Engineer*** **Dhaka, Bangladesh | September 2023 – Present**

Staff Information Bureau — Microcredit Regulatory Authority (MRA)

Technology Stack: Node.js, Hapi.js, Angular, Oracle, NG-ZORRO, Tailwind CSS

- Architected and independently developed the backend API layer for a centralized regulatory staff management system serving a user base of approximately one million, supporting staff records, employment history, inter-institutional transfers, and training information.
- Optimized database queries and transaction handling to improve data-processing efficiency, maintain consistency, and support reliable operations across a large-scale user base.
- Implemented transaction-safe record creation and update operations, strengthening data integrity and reducing the risk of duplicate-record conflicts during concurrent requests.
- Developed and enhanced Angular-based interfaces, contributing to responsive user experiences and efficient data management across institutional and regulatory workflows.
- Supported production rollout by diagnosing and resolving deployment-related and runtime issues, improving application stability and operational reliability.

Agent Licensing Online — Insurance Development and Regulatory Authority (IDRA)

Technology Stack: Node.js, Angular, PostgreSQL

- Took ownership of an existing insurance agent licensing platform, analyzing its codebase and business workflows to implement new features and improve existing application functionality.
- Investigated and resolved hidden defects across legacy modules, improving application reliability and ensuring accurate handling of agent licensing and renewal processes.
- Implemented CAPTCHA-based protection at critical entry points to mitigate automated bot activity and strengthen application security.
- Modified multi-level approval workflows to accommodate evolving business requirements and regulatory policy changes while preserving existing system functionality.
- Established an application and server monitoring solution using Grafana, Prometheus, and Loki to improve operational visibility, facilitate issue diagnosis, and support proactive production monitoring.
- Performed regular production server monitoring, troubleshooting, and maintenance to maintain system stability and minimize operational disruptions.

PROFESSIONAL EXPERIENCE (CONTINUED)

Celloscope Ltd. | ***Software Engineer*** **Dhaka, Bangladesh | September 2023 – Present**

Infrastructure Monitoring Facility — GIS-based monitoring of mobile towers

Technology Stack: NestJS, Angular, Node.js, PostgreSQL, n8n, RESTful APIs

- Architected and developed the complete backend REST API layer using NestJS for a nationwide GIS-based infrastructure monitoring platform for the telecom regulator (BTRC), tracking mobile towers and optical fiber networks.
- Engineered automated n8n workflows to continuously ingest real-time weather forecasts and river water-level data, enabling proactive flood monitoring and disaster risk assessment for telecom sites.
- Designed and optimized PostgreSQL database models and query workflows to efficiently handle high-volume geospatial data, tower health status, and alert triggers.
- Contributed to the frontend application in Angular, developing interactive GIS map visualizations and operational monitoring dashboards.

Integrated Microfinance Solution — Microcredit Regulatory Authority (MRA)

Technology Stack: Node.js, Hapi.js, Angular, PostgreSQL, NG-ZORRO

- Developed backend services supporting core microfinance operations, including member lifecycle management and Samity (center) administration, enabling structured and reliable operational data processing.
- Implemented role-based access control (RBAC) to support granular permissions across institutional roles, strengthening access governance and ensuring appropriate access to operational functions.
- Designed and integrated relational database operations with transaction management and query optimization to maintain data consistency across interconnected financial and operational records.
- Contributed to Angular-based management interfaces, developing data-driven forms, tables, and workflow components to streamline routine administrative operations.
- Supported the development of modular application components and business workflows to improve maintainability and accommodate diverse microfinance operational requirements.

Boijatra (E-commerce & Retail Platform)

Technology Stack: Next.js, Node.js, RESTful APIs, Online Payment Integration, Claude Code

- Built a full-stack e-commerce platform using Claude Code for AI-assisted development, leveraging Next.js for the frontend and developing the backend API layer to support core commerce operations.
- Utilized AI-assisted coding workflows to accelerate application development, implement features, and structure frontend and backend components across the application.
- Integrated online payment gateway functionality, including transaction verification and payment-status handling, to support secure and reliable checkout workflows.
- Developed administrative functionality for product catalog management, category organization, inventory tracking, and stock adjustments to support centralized retail operations.
- Implemented order lifecycle management, invoice generation, and sales reconciliation features to support order fulfillment and financial recordkeeping.

ENGINEERING PRACTICES & ARCHITECTURAL HIGHLIGHTS

- **Backend & API Architecture:** Hands-on RESTful service design using Node.js, Hapi.js, and Express.js, maintaining clear separation of concerns, route handling, input validation schemas, and business logic encapsulation.
- **Workflow Automation & Business Logic:** Engineered multi-stage business approval workflows, audit trails, and data partitioning mechanisms across enterprise platforms in regulatory and insurance domains.
- **Database Engineering & Consistency:** Experience writing and optimizing complex relational queries, schema migrations, and transaction management across PostgreSQL and Oracle production environments.
- **Production Reliability & Defect Triage:** Diagnosed and resolved critical production bugs in live nationwide systems, maintaining operational stability, data integrity, and high system availability.
- **Modern Frontend Integration:** Built responsive, accessible administrative dashboards and reactive interfaces using Next.js, Angular, NG-ZORRO UI framework, and Tailwind CSS.

EDUCATION

Khulna University — Bachelor of Science (B.Sc.) in Electronics and Communication Engineering **Khulna, Bangladesh | 2018 – 2023**

Govt. M.M. College — Higher Secondary Certificate (HSC) — Science **Jashore, Bangladesh | 2015 – 2017**